

Name:P Neeharika

Roll No: 2520090233

Session 1

Task1 – Linux Environment Setup

Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial

```
root@HP: ~/shellforge
root@HP:~# gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

root@HP:~# git --version
git version 2.43.0
root@HP:~# mkdir shellforge
root@HP:~# cd shellforge
root@HP:~/shellforge# pwd
/root/shellforge
root@HP:~/shellforge# |
```

Makefile

```
/root/shellforge
root@HP:~/shellforge# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: is subject to change. To configure the initial branch name to use in al
l
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/shellforge/.git/
root@HP:~/shellforge# git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
root@HP:~/shellforge#
```

Task 2:

Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing

```

root@HP:~# ls
demo1.c  fork_example  neeha    process.c  sri        xls.txt
demo2    fork_example.c neetu    session1.c states
demo2.c  nano.867.save process  shellforge states.c
root@HP:~# gcc session1.c -o session1
root@HP:~# ./session1
This is the parent process.
This is the child process.
Parent PID: 596
Child PID: 597
root@HP:~# |

```

```

Parent PID: 596
Child PID: 597
root@HP:~# ps -f
UID      PID     PPID  C  STIME TTY          TIME CMD
root      568      563  0  09:57 pts/0        00:00:00 -bash
root      601      568  50 10:01 pts/0        00:00:00 ps -f
root@HP:~# |

```

```

root@HP:~# ps -f
UID      PID     PPID  C  STIME TTY          TIME CMD
root      568      563  0  09:57 pts/0        00:00:00 -bash
root      601      568  50 10:01 pts/0        00:00:00 ps -f
root@HP:~# pstree -p
systemd(1)─agetty(218)
            └─cron(180)
               └─dbus-daemon(181)
                  └─init-systemd(Ub(2))─SessionLeader(562)─Relay(568)(563)─ba+
                     └─init(7)─{init}(8)
                        └─login(329)─bash(406)
                           └─{init-systemd(Ub)}(9)
                              └─rsyslogd(220)─{rsyslogd}(235)
                                     └─{rsyslogd}(236)
                                        └─{rsyslogd}(237)
                                           └─systemd(377)─(sd-pam)(380)
                                              └─systemd-journal(62)
                                                 └─systemd-logind(195)
                                                    └─systemd-resolve(121)
                                                       └─systemd-timesyn(122)─{systemd-timesyn}(169)
                                                          └─systemd-udev(114)─(udev-worker)(602)
                                                             └─(udev-worker)(603)
                                                                └─unattended-upgr(272)─{unattended-upgr}(278)
                                                                   └─wsl-pro-service(208)─{wsl-pro-service}(260)
                                                                      └─{wsl-pro-service}(261)
                                                                         └─{wsl-pro-service}(262)
                                                                            └─{wsl-pro-service}(263)
                                                                               └─{wsl-pro-service}(264)
                                                                                  └─{wsl-pro-service}(265)
                                                                                     └─{wsl-pro-service}(274)
root@HP:~#

```

Session 1 Observation:

The session provided practical experience in setting up the Linux development environment and understanding fundamental operating-system concepts related to processes. The first task established the required development tools and project structure, while the second task provided hands-on experience with process creation, program execution, parent-child relationships, process trees, and system-call tracing. Overall, the activities helped connect the theoretical concepts of operating systems with their practical implementation.

SESSION 2

Session 2 – Interactive Shell Main Loop and User Input Handling

Task 1: Create the main loop, display the prompt, read user input, handle exit conditions, design the control-flow diagram, and test the interactive

```
root@HP:~# nano session2.c
root@HP:~# gcc session2.c -o session2
root@HP:~# ./session2
mysHELL> hello
linux
operYou entered: hello
mysHELL> You entered: linux
mysHELL> ating system
exiYou entered: operating system
mysHELL> t
loop. root@HP:~# |

root@HP:~# nano session1_exec.c
root@HP:~#
root@HP:~# gcc session1_exec.c -o session1_exec
root@HP:~# ./session1_exec
total 160
-rw-r--r-- 1 root root    0 Aug  2 05:29 demo1.c
-rwxr-xr-x 1 root root 16088 Aug  2 05:44 demo2
-rw-r--r-- 1 root root   516 Aug  2 05:44 demo2.c
-rwxr-xr-x 1 root root 16280 Aug  1 04:16 fork_example
-rw-r--r-- 1 root root   591 Aug  1 04:15 fork_example.c
-rw----- 1 root root    27 Aug  1 03:52 nano.867.save
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neeha
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neetu
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 process
-rw-r--r-- 1 root root   706 Aug  8 03:16 process.c
-rwxr-xr-x 1 root root 16128 Aug 20 09:59 session1
-rw-r--r-- 1 root root   441 Aug 20 09:58 session1.c
-rwxr-xr-x 1 root root 16016 Aug 20 10:13 session1_exec
-rw-r--r-- 1 root root   215 Aug 20 10:13 session1_exec.c
-rwxr-xr-x 1 root root 16184 Aug 20 10:10 session2
-rw-r--r-- 1 root root   349 Aug 20 10:10 session2.c
drwxr-xr-x 3 root root  4096 Aug 20 09:57 shellforge
drwxr-xr-x 2 root root  4096 Jul 31 18:16 sri
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 states
-rw-r--r-- 1 root root   576 Aug  8 03:17 states.c
-rw-r--r-- 1 root root    0 Jul 31 18:28 xls.txt
root@HP:~# |

root@HP:~# ./session2
mysHELL>
You entered:
mysHELL> exit
root@HP:~# |
```

Task 2: Capture keyboard input, handle Backspace, process the Enter key, manage the input buffer, support multi-character commands, and test user interaction.

```
root@HP:~# nano session2.c
root@HP:~# gcc session2.c -o session2
root@HP:~# ./session2
myshell> hellow
You entered: hellow
myshell> hello world
You entered: hello world
myshell>
You entered:
myshell> exit
root@HP:~# |

myshell> exit
root@HP:~# nano session2_keyboard.c
root@HP:~# gcc session2_keyboard.c -o session2_keyboard
root@HP:~# gcc session2_keyboard.c -o session2_keyboard
root@HP:~# ./session2_keyboard
myshell> hello
hello
You entered: hello
root@HP:~# ./session2_keyboard
myshell> helllo
helllo
You entered: helllo
root@HP:~# |
```

Session 2 Observation:

The session focused on developing the interactive input functionality of the shell. The first task established the main shell loop, prompt, input reading, and exit mechanism, while the second task concentrated on keyboard-level input handling and buffer management. Through practical testing, the behavior of the shell during user interaction was observed and verified. The session provided a clear understanding of how an interactive shell receives, processes, and manages user input.